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Out-of-Distribution Detection Using an Ensemble of Self Supervised Leave-out Classifiers

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Abstract. As deep learning methods form a critical part in commercially important applications such as autonomous driving and medical diagnostics, it is important to reliably detect out-of-distribution (OOD) inputs while employing these algorithms. In this work, we propose an OOD detection algorithm which comprises of an ensemble of classifiers. We train each classifier in a self-supervised manner by leaving out a random subset of training data as OOD data and the rest as in-distribution (ID) data. We propose a novel margin-based loss over the softmax output which seeks to maintain at least a margin m between the average entropy of the OOD and in-distribution samples. In conjunction with the standard cross-entropy loss, we minimize the novel loss to train an ensemble of classifiers. We also propose a novel method to combine the outputs of the ensemble of classifiers to obtain OOD detection score and class prediction. Overall, our method convincingly outperforms Hendrycks *et al.* [7] and the current state-of-the-art ODIN [13] on several OOD detection benchmarks.

Keywords: anomaly detection · out-of-distribution

1 Introduction

Deep learning has significantly improved the performance of machine learning systems in fields such as computer vision, natural language processing, and speech. In turn, these algorithms are integral in commercial applications such as autonomous driving, medical diagnosis, and web search. In these applications, it is critical to detect sensor failures, unusual environments, novel biological phenomena, and cyber attacks. To accomplish this, systems must be capable of detecting when inputs are anomalous or out-of-distribution (OOD). In this work, we propose an out-of-distribution detection method for deep neural networks and demonstrate its performance across several out-of-distribution classification tasks on the state-of-the-art deep neural networks such as DenseNet[8] and Wide ResNet(WRN)[22].

⋆ Equal contribution. Work done when the authors were working at Intel labs.

We propose a novel margin-based loss term, added to cross-entropy loss over in-distribution samples, which maintains a margin of at least m between the average entropy of OOD and ID samples respectively. We propose an ensemble of K leave-out classifiers for OOD detection. The training dataset with N classes is partitioned into K subsets such that the classes of each partition are mutually exclusive with respect to each other. Each classifier samples one of the K subsets without replacement as *out-of-distribution training data* and the rest of the $K-1$ subsets as *in-distribution training data*. We also propose a new OOD detection score which combines both softmax prediction score and entropy with temperature scaling [13]. We demonstrate the efficacy of our method on standard benchmarks proposed in ODIN [13] and *outperform them*. Our contributions are (i) proposing a novel loss for OOD detection, (ii) demonstrating a self-supervised OOD detection method, and (iii) moving the state-of-the-art by outperforming the current best methods.

The rest of the paper is organized as follows. Section 2 describes the previous work on the OOD detection. Section 3 describes our method in detail. Section 4 describes various evaluation metrics to measure the performance of OOD detection algorithms. The ablation results of various design choices and hyper-parameters are also presented. We then compare our method against the recently proposed ODIN algorithm [13] and demonstrate that it outperforms it on various OOD detection benchmarks. Finally, section 5 discusses observations about our method, future directions and conclusions.

2 Related Work

Traditionally, based on the availability of the data labels, OOD detection methods can be categorized into supervised [16], semi-supervised [4] and unsupervised methods [15], [3]. All these classes of methods have access to the OOD data while training but differ in access to labels. It is assumed that the classifier has labels for normal as well as OOD classes during training for supervised OOD detection, while labels for only the normal classes are available in case of semi-supervised methods, and no labels are provided for unsupervised OOD detection methods which typically rely on the fact that anomalies occur in much less frequency than normal data. Our method is able to detect anomalies in test OOD datasets the very first time it encounters them during testing. We use one OOD dataset as validation set to search for hyper-parameters.

Notable OOD detection algorithms which work in the same setting as ours are isolation forests [14], Hendrycks and Gimpel [7], ODIN [13] and Lee *et.al,* [12]. The work reported in isolation forests [14] exploits the fact that anomalies are scarce and different and while constructing the isolation tree, it is observed that the anomalous samples appear close to the root of the tree. These anomalies are then identified by measuring the length of the path from the root to a terminating node; the closer a node is to the root, the higher is its chance of representing an OOD. Hendrycks and Gimpel [7] is based on the observation that prediction probability of incorrect and out-of-distribution samples tends to be lower than the prediction

probability of correct samples. Lee *et al.* modify the formulation of generative adversarial networks [6] to generate OOD samples for the given in-distribution. They achieve this by simultaneously training GAN [6] and standard supervised neural network. The joint loss consists of individual losses and an additional connecting term which reduces the KL divergence between the generated sample's softmax distribution and the uniform distribution.

Another set of related works are open set classification methods [17], [18], [19], [1], [2]. Scheirer *et.al,* [19] introduces and formalizes "open space risk" which intuitively is the risk associated with labeling those areas in the output feature space as positive where there is no density support from the training data. Thus the approximation to the ideal risk is defined as a linear combination of "open space risk" and the standard "empirical risk". Bendale and Boult [1] extend the definition of "open set risk" to open world recognition where the unknown samples are not static set. The open world recognition defines a *multi-class open set recognition function*, a *labeling process* and an *incremental learning function*. The *multi-class open set recognition function* detects novel classes which are labeled using the *labeling process* and finally are fed to *incremental learning function* which updates the model. The OSDN [2] work proposes openMax function which extends the softmax function by adding an additional unknown class to the classification layer. The value for unknown class is computed by taking the weighted average of all other classes. The weights are obtained from Weibull distribution learnt over the pairwise distances between penultimate activation vectors (AV) of the top farthest correctly classified samples. For an OOD test sample these weights will be high while for an in-distribution sample these scores will be low. The final activation vector is re-normalized using softmax function.

The current state-of-the-art is ODIN [13] which proposes to increase the difference between the maximum softmax scores of in-distribution and OOD samples by (i) calibrating the softmax scores by scaling the logits that feed into softmax by a large constant (referred to as temperature) and (ii) pre-processing the input by perturbing it with the loss gradient. ODIN [13] demonstrated that at high temperature values, the softmax score for the predicted class is proportional to the relative difference between largest unnormalized output (logit) and the remaining outputs (logits). Moreover, they empirically showed that the difference between the largest logit and the remaining logits is higher for the in-distribution images than for the out-of-distribution images. Thus temperature scaling pushes the softmax scores of in- and out-of-distribution images further apart when compared to plain softmax. Perturbing the input image through gradient ascent w.r.t to the score of predicted label was demonstrated [13] to have stronger effect on the in- distribution images than that on out-of-distribution images, thereby, further pushing apart the softmax scores of in- and out-of-distribution images. We leverage the effectiveness of both these methods. The proposed method outperforms all the above methods by considerable margins.